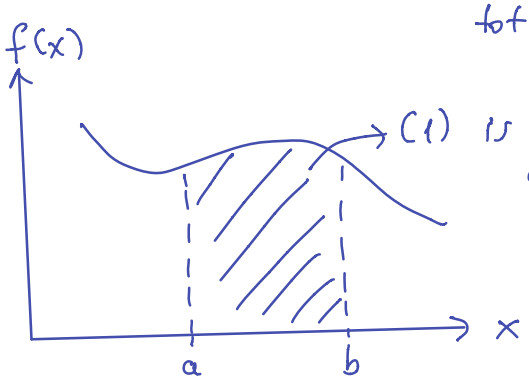
Chapter 19 – Numerical Integration

Integration: to bring together; to unite

Definite integration : $I = \int_a^b f(x)\,dx$ (1)

$\int_a^b f(x)\,dx$ → total value of $f(x)\,dx$ over $x = a$ to b.



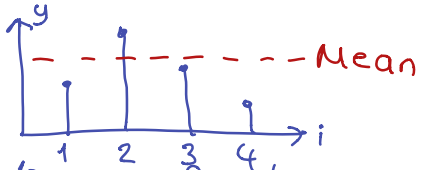
(1) is equivalent to the area under the curve

Analogy between integration and summation

⇒ Mean of n discrete data pts:

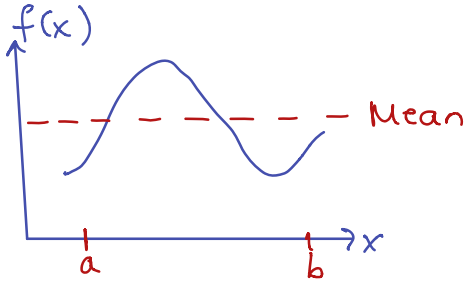
$$\text{Mean} = \frac{\sum_{i=1}^{n} y_i}{n}$$

y_i → individual measurements



⇒ Mean or average of a continuous function $y = f(x)$

$$\text{Mean} = \frac{\int_a^b f(x)\,dx}{b-a}$$



* Integral may be evaluated over a line, an area, or a volume

$$\text{Mass} = \iiint_V \underbrace{c(x,y,z)}_{\text{concentration}}\,dx\,dy\,dz \rightarrow \text{volume integral}$$

$$= \iiint_V c(V)\,dV$$

$$\text{Flux} = \iint_A f(x,y)\,dA \rightsquigarrow \text{Area integral}$$

* If functions to be integrated are simple ⇒ evaluate analytically. (exact result)

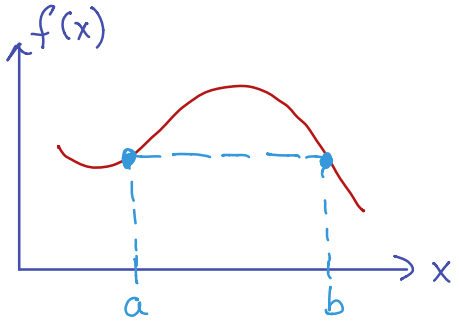
* If no analytical integration can be found ⇒ numerical techniques to obtain approximate value.

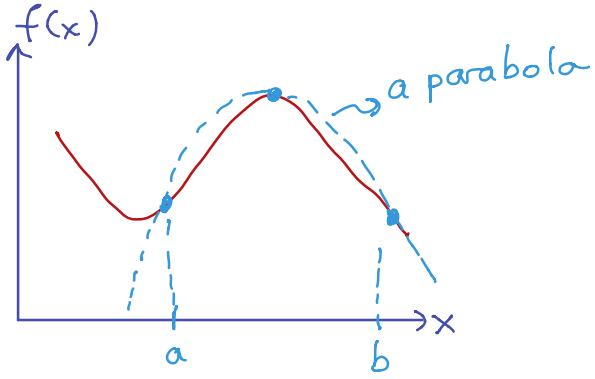
Newton-Cotes Formulas

→ Most common technique

→ Idea: replace a complicated function or tabulated value with a polynomial that is easy to integrate

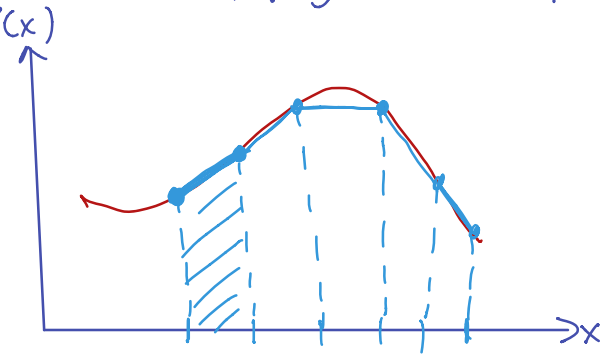
$$I = \int_a^b f(x)\,dx \cong \int_a^b f_n(x)\,dx \quad , f_n(x) = a_0 + a_1 x + \cdots a_n x^n$$



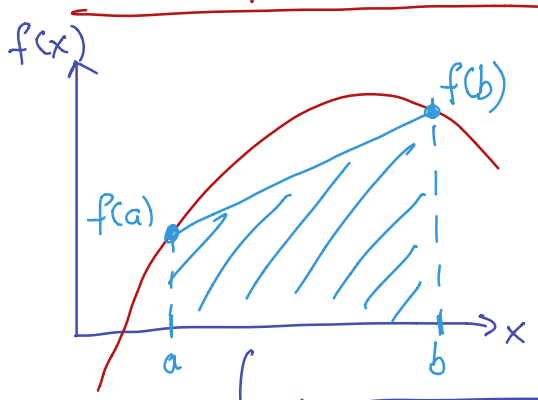


first-order polynomial (line)

(*) Use a series of polynomials applied piecewise to the funct.



The Trapezoidal Rule



$$I = \int_a^b f(x)\,dx \cong \int_a^b f_n(x)\,dx$$

↓ 1st order poly.

$$= \int_a^b \left[f(a) + \underbrace{\frac{f(b)-f(a)}{b-a}}_{\text{slope}} (x-a) \right] dx$$

Area of the trapezoid

$$\boxed{I = (b-a)\frac{f(a)+f(b)}{2}} \quad (1)$$

= width × average height

Error of the Trapezoidal Rule:

$$\boxed{E_t = -\frac{1}{12} f''(\xi)(b-a)^3}$$ ξ : between a and b

(if $f''(\xi)=0$ (the func. is linear) ⇒ the error is zero)

Ex Numerically integrate from $a=0$ to $b=0.8$

$f(x) = 0.2 + 25x - 200x^2 + 675x^3 - 900x^4 + 400x^5$

(Exact value: 1.640533)

Sol'n $f(a) = f(0) = 0.2$

$f(b) = f(0.8) = 0.232$

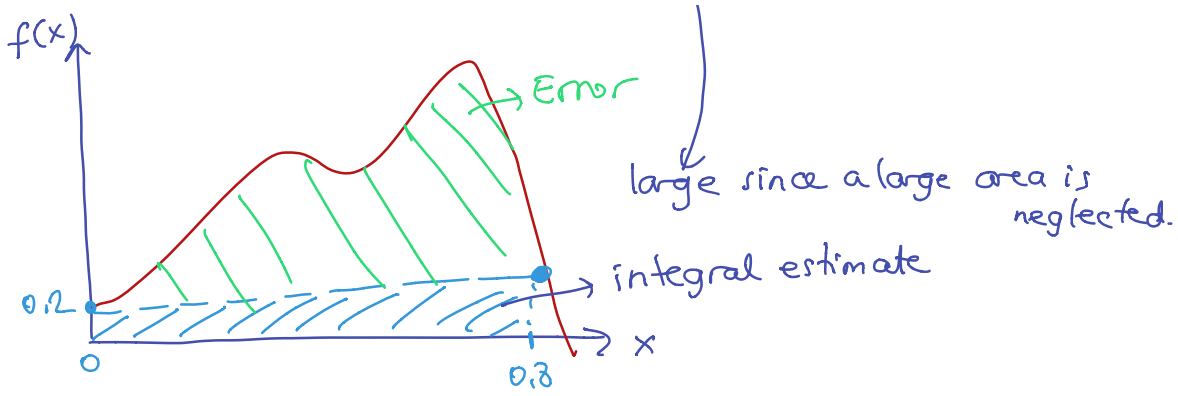
} insert into (1)

$$\Rightarrow I = (0.8-0)\frac{0.2+0.232}{2}$$

$$I = 0.1728$$

$E_t = 1.640533 - 0.1728 = 1.467733$ (error)

$$\varepsilon_t = \frac{1.467733}{1.640533} \times 100 = 89.5\%$$ (relative percent error)



The Composite Trapezoidal Rule

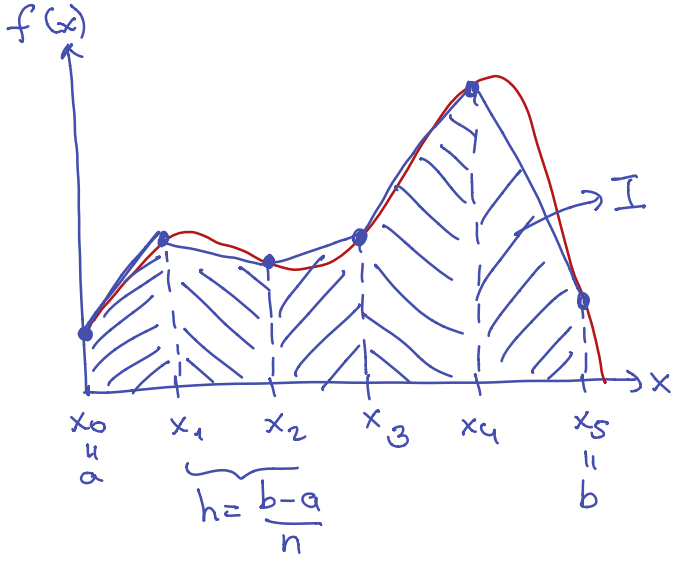
To improve accuracy ⇒ divide the interval from a to b into segments.

$(x_0, x_1, x_2 \cdots x_n)$: (n+1) equally spaced ptrs.

$x_0 = a$, $x_n = b$

$$h = \frac{b-a}{n}$$

$$I = \int_{x_0}^{x_1} f(x)\,dx + \int_{x_1}^{x_2} f(x)\,dx + \cdots + \int_{x_{n-1}}^{x_n} f(x)\,dx$$



Substitute trapezoidal rule for each integral:

$$I = h\frac{f(x_0)+f(x_1)}{2} + h\frac{f(x_1)+f(x_2)}{2} + \cdots + h\frac{f(x_{n-1})+f(x_n)}{2}$$

Group terms:

$$\boxed{I = \frac{h}{2}\left[f(x_0) + 2\sum_{i=1}^{n-1} f(x_i) + f(x_n)\right]}$$

$$I = \underbrace{(b-a)}_{\text{width}} \underbrace{\frac{f(x_0) + 2\sum_{i=1}^{n-1} f(x_i) + f(x_n)}{2n}}_{\text{average height}}$$

= weighted average of the function values

Error: $E_t = -\frac{(b-a)^3}{12n^3}\sum_{i=1}^{n} f''(\xi_i)$ ↳ 2nd der. of apt. located in segment i.

to simplify

$$\bar{f}'' = \frac{\sum_{i=1}^{n} f''(\xi_i)}{n}$$

$$\boxed{E_a = -\frac{(b-a)^3}{12n^2}\bar{f}''}$$

if n (# segments) is doubled, error will decrease by a factor of 4.

Ex Numerically integrate from $a=0$ to $b=0.8$

$f(x) = 0.2 + 25x - 200x^2 + 675x^3 - 900x^4 + 400x^5$

(Exact value: 1.640533)

Sol^n For $n=2$ $\left(h=\frac{0.8-0}{2}=0.4\right)$

$f(0)=0.2$, $f(0.4)=2.456$, $f(0.8)=0.232$

$$I=(b-a)\frac{f(x_0)+2\sum_{i=1}^{n-1}f(x_i)+f(x_n)}{2n}$$

$$=(0.8)\frac{0.2+2(2.456)+0.232}{2(2)}=1.0688$$

$E_t = 1.640533 - 1.0688 = 0.57173 \rightarrow \varepsilon_t = \underline{34.9\%}$

↓ true value

Simpson's Ruler

Improve the accuracy: use higher-order polynomial to connect the pts.

$$I=\int_a^b f(x)dx \cong \int_a^b f_n(x)dx$$

↓ 2nd order polynomial

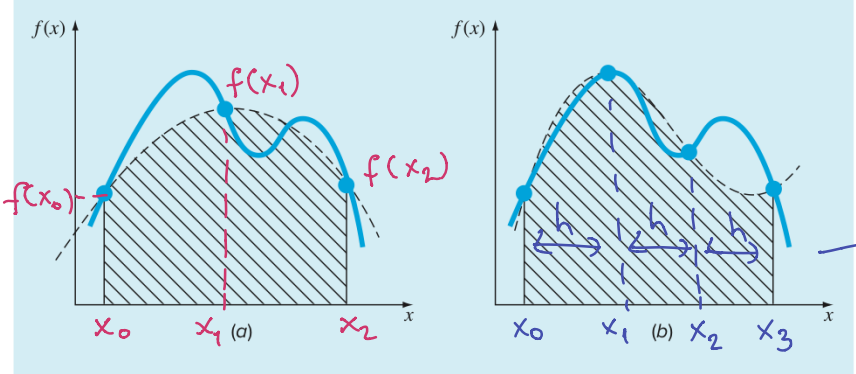
, Let $a=x_0$, $b=x_2$, x_1 btwn $[x_0,x_2]$

Three pts: $(x_0,f(x_0)),(x_1,f(x_1)),(x_2,f(x_2))$

$$I=\int_{x_0}^{x_2}\left[\frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)}f(x_0)+\frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)}f(x_1)+\frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)}f(x_2)\right]dx$$

$f_2(x)$: 2nd order polynomial

FIGURE 19.11
(*a*) Graphical depiction of Simpson's 1/3 rule: It consists of taking the area under a parabola connecting three points. (*b*) Graphical depiction of Simpson's 3/8 rule: It consists of taking the area under a cubic equation connecting four points.



$\rightarrow h = \frac{b-a}{3}$

$= \frac{x_3 - x_0}{3}$

After algebraic manipulations:

$$I = \frac{h}{3}\left[f(x_0) + 4f(x_1) + f(x_2)\right]$$

, $a = x_0$ and $b = x_2$ where $h = \frac{b-a}{2}$

x_1 : pt. midway between a and b.

Simpson's $\frac{1}{3}$ Rule

(2) $$I = \underbrace{(b-a)}_{2h}\ \frac{f(x_0) + 4f(x_1) + f(x_2)}{6}$$

Truncation Error $E_t = -\frac{1}{90} h^5 f^{(4)}(\xi)$, $h = \frac{b-a}{2}$

$$E_t = -\frac{(b-a)^5}{2880} f^{(4)}(\xi)$$, ξ is btwn a and b.

* Simpson's rule is more accurate than trapezoidal Rule
* " " yields exact results for cubic polynomials even though it is derived from a parabola.

Ex Numerically integrate from $a=0$ to $b=0.8$

$f(x) = 0.2+25x-200x^2+675x^3-900x^4+400x^5$

(Exact value : 1.640533)

Sol^n use Simpson's $\frac{1}{3}$ Rule

$n=2 \quad (h=0.4)$

$f(0)=0.2 \quad , \quad f(0.4)=2.456 \quad , \quad f(0.8)=0.232$

from (2) $\Rightarrow I=(b-a)\dfrac{f(x_0)+4f(x_1)+f(x_2)}{6}$

$$I=(0.8)\frac{0.2+4(2.456)+0.232}{6}=1.367467$$

$\varepsilon_t=16.6\%$ $\rightarrow$ five times more accurate than a single application of trapezoidal rule

The Composite Simpson's $\frac{1}{3}$ Rule

$$I=\underbrace{\int_{x_0}^{x_2} f(x)dx}_{x_0}+\underbrace{\int_{x_2}^{x_4} f(x)dx}_{x_2}+\cdots+\int_{x_{n-2}}^{x_n} f(x)dx$$

Substitute Simpson's $\frac{1}{3}$ Rule for each interval

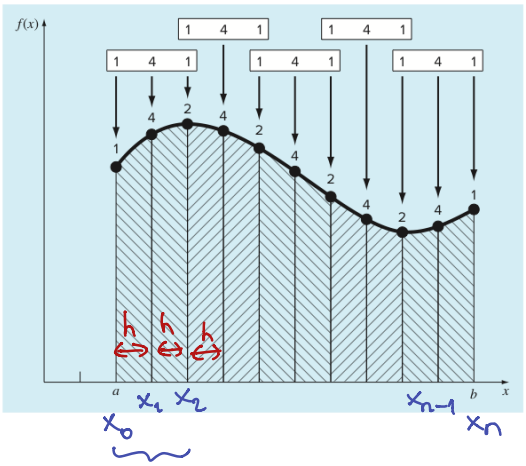
$$I=2h\frac{f(x_0)+4f(x_1)+f(x_2)}{6}+2h\frac{f(x_2)+4f(x_3)+f(x_4)}{6}+\cdots\cdots$$

$$+2h\frac{f(x_{n-2})+4f(x_{n-1})+f(x_n)}{6}$$

Group terms:

$$I=(b-a)\frac{f(x_0)+4\sum_{i=1,3,5\cdots}^{n-1} f(x_i)+2\sum_{j=2,4,6\cdots}^{n-2} f(x_j)+f(x_n)}{3n} \quad (3)$$

FIGURE 19.12
Composite Simpson's 1/3 rule. The relative weights are depicted above the function values. Note that the method can be employed only if the number of segments is even.



Error Estimate

$$E_a = -\frac{(b-a)^5}{180n^4}\overline{f^{(4)}}$$ ←

EX Same example as before

Use $n=4$ $\quad h = \frac{0.8-0}{4} = 0.2$

$f(0) = 0.2$, $f(0.2) = 1.288$ ($n=1$), $f(0.4) = 2.456$ ($n=2$)

$f(0.6) = 3.464$ ($n=3$)

$f(0.8) = 0.232$ ($n=4$)

use (3)

$$I = 0.8\,\frac{0.2 + 4(1.288 + 3.464) + 2(2.456) + 0.232}{3(4)} = 1.623467$$

$\varepsilon_t = 1.04\%$

* Composite Simpson's $\frac{1}{3}$ rule is limited to cases where the values are equispaced.

Simpson's $\frac{3}{8}$ Rule

Fit a 3rd order Lagrange polynomial to 4 pts. and integrate

$$I = \frac{3h}{8}\left[f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)\right], \quad h = \frac{b-a}{3}$$

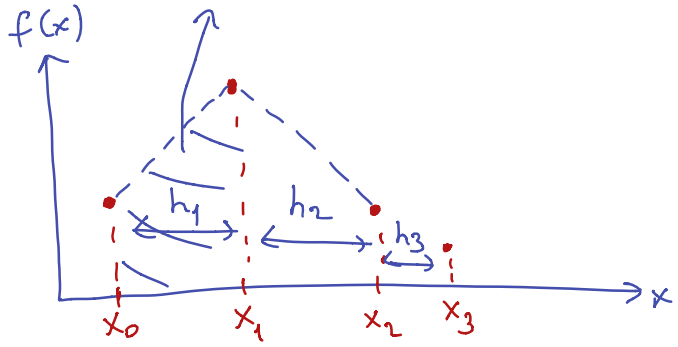
$$E_t = -\frac{3}{80} h^5 f^{(4)}(\xi) \quad \text{(Error)}$$

Integration with Unequal Segments

Experimental data may have unequal sized segments.

Apply trapezoidal rule to each segment and sum the results.

$$I = \underbrace{h_1 \frac{f(x_0)+f(x_1)}{2}} + h_2 \frac{f(x_1)+f(x_2)}{2} + \cdots + h_n \frac{f(x_{n-1})+f(x_n)}{2}$$



MATLAB Functions:

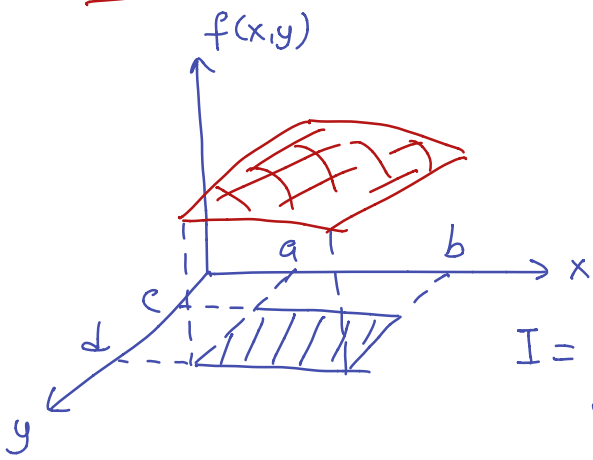
trapz :

cumtrapz :

z = trapz (x, y)

- y → dependent variable
- x → indep. var

Multiple Integrals



- Evaluate the integral in one dimension first
- Integrate the result in the second dimension.

$$I = \int_c^d \left(\int_a^b f(x,y)\,dx \right) dy$$

$$= \int_a^b \left(\int_c^d f(x,y)\,dy \right) dx$$

$$\bar{f} = \frac{I}{(b-a)(d-c)}$$: average value of $f(x,y)$

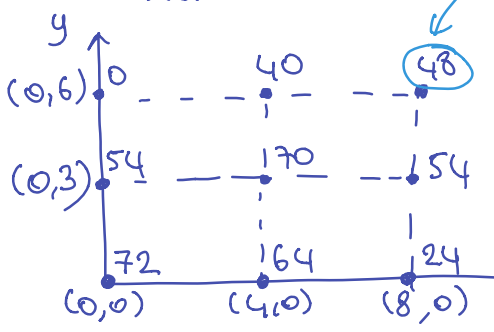
Ex Temperature of a rectangular heated plane is described by:

$$T(x,y) = 2xy + 2x - x^2 - 2y^2 + 72$$

If the plane is 8 m long (x dimension) and 6m wide (y dimension), compute the average.

Sol^n (Analytical solution: 58.66667)

Two-segment application of trapezoidal rule in each dimension



$x=8, y=6, T(x,y)$

integrate w.r.t x

fix $y=6$ → $(8-0)$ $\quad \frac{0+2(40)+48}{4} = 256$

fix $y=3$ → $(8-0)$ $\quad \frac{54+2(70)+54}{4} = 496$

fix $y=0$ → $(8-0)$ $\quad \frac{72+2(64)+24}{4} = 448$

(Integrate w.r.t y) ←- - - - -

$$2544 = (6-0)\frac{256+2(496)+448}{4}$$

Divide by area $\frac{2544}{6\times 8} = 53$ degrees.

Solve by single segment Simpson's Rule: $\frac{2816}{48} = 58.66667$

MATLAB Functions:

dblquad

triplequad

```
q = dblquad(fun, xmin, xmax, ymin, ymax, tol)
```

tol: default tolerance 10^{-6}

```
>> q = dblquad(@(x,y) 2*x*y + 2*x - x.^2 - 2*y.^2 + 72, 0, 8, 0, 6)

q = 2816
```